1 Building the model

We study how the geometry of a two-dimensional elastic object and its supports affect its response to applied forces. Our unknown is the displacement $\mathbf{u}(x, y)$, which tells us how far each point moves horizontally and vertically. Differences in movement between nearby points describe local stretching and shearing, called strain. The material resists these changes through internal forces per unit area, called stress. We assume small deformations and a linear relation between strain and stress, leaving the material choice to the individual cases [1]. Computing this response leads to a large sparse linear system, so we also ask which solvers suit it and how their cost grows as the mesh is refined.

We look for the shape after the object has settled under the load. Forces must then balance in both directions throughout the material:

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + g_x = 0, \qquad \frac{\partial \sigma_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + g_y = 0. \tag{1}$$

Here the σ terms are stress components and g_x, g_y are applied forces per unit volume. Since stress depends on strain, and strain depends on displacement, Eq. (1) gives equations for $\mathbf{u}$. We also specify where forces act on the boundary and which movements the supports prevent.

To make this a finite problem, we divide the region into filled triangles as shown in Fig. 1, following its boundary and leaving any holes empty [2]. Their vertices are the mesh nodes, each with two unknown displacement components. Inside each triangle, displacement varies linearly between its three vertex values. Adjacent triangles share these values, so their displacements agree along the common edge. This represents movement throughout the solid; the triangle edges are not physical bars. Smaller triangles allow us to capture more variation in the deformation.

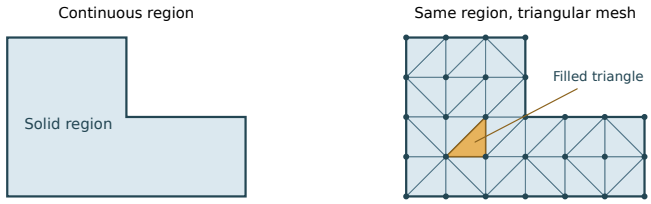


Figure 1: Triangulation of an illustrative region. The filled triangles cover the same solid area and share nodes along their edges. Displacement is approximated throughout each triangle using its three vertex values. The outline is an example, not a selected project geometry.

The finite element method applies Eq. (1) by balancing internal and external work, meaning forces acting through small allowed displacements. Using the triangle approximation and adding contributions at shared nodes gives

$$Kd = f + r, \tag{2}$$

where d collects node displacements, f the applied forces and r the forces exerted by supports. The stiffness matrix K describes how the mesh and material resist displacement. Most entries are zero because only nodes in a common triangle interact directly.

Finally, we separate unknown displacements (F) from values prescribed at supports (C). The free rows of Eq. (2) become

$$Ax = b. \tag{3}$$

Here $A = K_{FF}$, $x = d_F$ and $b = f_F - K_{FC}d_C$; for fixed supports, $d_C = 0$ and $b = f_F$. Supports must prevent each separate piece from freely sliding or turning. Solving Eq. (3) then gives the displacements used to draw the deformed object.

References

[1] Anders Logg and Hans Petter Langtangen, *The equations of linear elasticity*, FEniCSx tutorial, maintained by Jørgen Schartum Dokken, Accessed 10 November 2026.

[2] Jonathan Richard Shewchuk, *Triangle: Generating Delaunay triangulations*, Carnegie Mellon University, Triangle documentation, Accessed 10 November 2026.